

# NINA RADUNER

NEUROSCIENTIST

Department of Child and Adolescent Psychiatry | Neumünsterallee 9 | CH-8032 Zurich  
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EDUCATION

|                 |                                                                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10/21 - present | <b>DOCTORAL CANDIDATE, DEPARTMENT OF PSYCHOLOGY</b><br><b>University of Zurich, Zurich, Switzerland</b><br><b>Supervisor: Prof. S. Brem</b><br>investigating multisensory learning in healthy children<br>planned graduation: 05/25 |
| 08/19 - 07/21   | <b>M. SC. PSYCHOLOGY, DEPARTMENT OF PSYCHOLOGY</b><br><b>University of Zurich, Zurich, Switzerland</b><br>ERP study of a visual Lexical Decision Task                                                                               |
| 08/15 - 07/19   | <b>B. SC. PSYCHOLOGY, DEPARTMENT OF PSYCHOLOGY</b><br><b>University of Zurich, Zurich, Switzerland</b>                                                                                                                              |

FELLOWSHIPS

|                 |                                                                                                                                                              |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10/21 - Present | <b>INTERNATIONAL PHD PROGRAM IN NEUROSCIENCE (ZNZ)</b><br><b>Life Science Zurich Graduate School</b><br>Joint program by University of Zurich and ETH Zurich |
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EXPERIENCE

|                                |                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10/21 - present                | <b>Doctoral research associate in Developmental Cognitive Neuroscience Group at Department of Child and Adolescent Psychiatry, University of Zurich, Switzerland (Prof. S. Brem)</b><br>writing ethics application, planning of measurements, recruitment, supervision, data collection (fMRI, behavioural), data analysis (behavioural, computational modelling, fMRI) |
| 04/19 - 08/19                  | <b>Night assistant at Sleep Laboratory Fluntern, Zurich, Switzerland</b><br>Patient intake, medical history interviews, electrode application and patient preparation for sleep study                                                                                                                                                                                   |
| 09/20 - 02/21<br>09/19 - 02/20 | <b>Research tutorage at Department of Neuropsychology, University of Zurich, Switzerland (Prof. L. Jäncke)</b><br>EEG and psychometric testing                                                                                                                                                                                                                          |
| 04/19 - 08/19                  | <b>Research internship at Department of Neuropsychology, University of Zurich, Switzerland (Prof. L. Jäncke)</b><br>planning and conduction of EEG and psychometric testing                                                                                                                                                                                             |
| 02/19 - 04/19                  | <b>Research internship at Sport Psychology and Movement Sciences, Stuttgart, (Prof. N. Schott)</b><br>analysing mental training data, data collection in nursing homes                                                                                                                                                                                                  |

STATEMENT

I was first introduced to computational modelling during my Master's in Neuropsychology at the University of Zurich. My PhD further developed my expertise in reinforcement learning, neuroimaging, and behavioural analysis, focusing on multisensory (audio-visual and tactile-visual) learning. I have a strong passion for the methodological aspects of neuroscience and am eager to apply my skills in computational modelling and neuroimaging.

LANGUAGES

|              |       |
|--------------|-------|
| Swiss German | ●●●●● |
| German       | ●●●●● |
| English      | ●●●●○ |
| French       | ●●○○○ |

SKILLS

|             |       |
|-------------|-------|
| R, R-Studio | ●●●●● |
| Matlab      | ●●●●● |
| SPM12       | ●●●●○ |
| Python      | ●●●○○ |
| Nilearn     | ●●○○○ |
| EEGLAB      | ●●○○○ |
| Automagic   | ●●○○○ |
| ERPLAB      | ●●○○○ |
| FieldTrip   | ●●○○○ |

# CURRICULUM VITAE

## GRANTS

|                |                                                                                                                                                                                                                                     |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03/25<br>03/24 | <b>PhF Peer Mentoring; main applicant for the project "Brainfood - exchange of methods and knowledge for early career human neuroimaging researchers"</b><br>awarded amount for 2024: CHF 4'360; amount awarded for 2025: CHF 3'500 |
| 12/22          | <b>GRC Peer Mentoring; main applicant for the project "Brainfood - exchange of methods and knowledge for early career human neuroimaging researchers"</b><br>Nr: 2022_Q3_PM_008; awarded amount for 1 year: CHF 4'420               |

## TEACHING & MENTORSHIP

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|-------------|---------------------------------------------------------------------------------|
| HS22        | Co-organizer and -instructor of a workshop - "fMRI data analyses for beginners" |
| HS22 - FS24 | Supervising master students and interns: S. Gubler, M.A. Steuer, Y. Cao         |

## EXTRACURRICULAR ACTIVITIES

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|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01/23 - present | <b>Co-organizer of a GRC funded peer mentoring group "Brainfood"</b><br>Securing funding, co-organizing monthly lunch seminars, workshops, and guest lectures |
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## PUBLICATIONS

**Raduner, N**, Providoli, C, Di Pietro, S, Schneebeli, M, Karipidis, I. I, Casimiro, E, Bedi S, Von Rhein M, Raschle NM, Ruff CC, & Brem, S. (2025). Growing Minds, Integrating Senses: Neural and Computational Insights into Age-related Changes in Audio-Visual and Tactile-Visual Learning in Children. *BioRxiv*, 2025.03.04.639748. <https://doi.org/10.1101/2025.03.04.639748>

## PEER REVIEWED CONFERENCE PROCEEDINGS

Providoli C, Di Pietro SV, **Raduner N**, Ehrhardt N, Scatolin S, Karipidis II, Haugg A, Ruff C, von Rhein M, Brem S, Raschle NM (2024) Neural correlates of movie-based audiovisual. Flux Annual Meeting, September 2024, Baltimore (MD)

Nohl A, Providoli C, **Raduner N**, Frei N, Di Pietro SV, Lutz C, Karipidis II, Brem S (2024) Exploring the Visual Word Form Area and Fusiform Face Area Activation during reading acquisition. ZNZ Symposium, September 2024, Zurich (CH)

Providoli C, Wombacher R, Di Pietro SV, **Raduner N**, Ehrhardt N., Scatolin S, Karipidis II, Haugg A, Ruff C, von Rhein M, Brem S, Raschle NM (2024) How children and adults differ in functional connectivity during audiovisual processing in a movie paradigm. ZNZ Symposium, September 2024, Zurich (CH)

**Raduner N**, Schneebeli M, Providoli C, Di Pietro SV, Bedi S, Casimiro E, Raschle NM, von Rhein M, Ruff C, Brem S (2024) Neural Correlates of Prediction Errors During Multisensory Learning in Children. OHBM Annual Meeting, June 2024, Seoul (KOR)

**Raduner N**, Providoli C, Di Pietro SV, Bedi S, Casimiro E, Karipidis II, Schneebeli M, Raschle NM, Ruff C, von Rhein M, Brem S (2024) Brain Activation During Multisensory Feedback Learning in Children. Swiss Society for Early Childhood Research Annual Meeting, January 2024, St. Gallen (CH)

Providoli C, **Raduner N**, Di Pietro SV, Scatolin S, Strickler L, Ehrhardt N, Borbás R, Dimanova P, Karipidis II, Haugg A, Ruff C, von Rhein M, Brem S, Raschle NM (2024) Narration Processing During a Movie Watching Paradigm in the Developing Brain. Swiss Society for Early Childhood Research Annual Meeting, January 2024, St. Gallen (CH)

**Raduner N**, Providoli C, Di Pietro SV, Bedi S, Casimiro E, Ehrhardt N, Gubler S, Karipidis II, Schneebeli M, von Rhein M, Raschle NM, Ruff C, Brem S (2023) Prediction errors modulate brain responses during multisensory feedback learning in adults. Society for the Neurobiology of Language Annual Meeting 2023, Marseille (FR)

**Raduner N**, Providoli C, Di Pietro SV, Bedi S, Casimiro E, Ehrhardt N, Gubler S, Karipidis II, Schneebeli M, von Rhein M, Raschle NM, Ruff C, Brem S (2023) Brain activation during audio-visual and tactile-visual feedback learning in adults. ZNZ Symposium, September 2023, Zurich (CH)

**Raduner N**, Ehrhardt N, Providoli C, Di Pietro SV, Bedi S, Casimiro E, Chyl K, Karipidis II, Borbás R, Dimanova P, Jednoróg K, Ruff C, von Rhein M, Raschle NM, Brem S (2022) Brain activation during audio-visual and tactile-visual feedback learning in adults. Burghölzli Psychiatry Meeting 2022, Zurich (CH)